

Control Unit Architecture & Hardwired/Microprogrammed Control

Q. What is the role of the Control Unit? Differentiate between Hardwired Control Unit and Microprogrammed Control Unit.

> **Definition to Remember**

1. Hardwired Control Unit

The control logic is implemented physically using digital hardware circuits (logic gates, flip-flops, decoders).

2. Microprogrammed (Soft-Wired) Control Unit

The control signals are generated by executing a tiny, low-level program (a **Microprogram**) stored in a special, fast memory called **Control Memory (ROM)**.

3. Key Differences

Feature	Hardwired CU	Microprogrammed CU
Control Logic	Implemented physically using digital hardware circuits.	Generated by executing a microprogram stored in Control Memory (ROM).
Flexibility	Not flexible; changes require physical redesign.	Flexible; changes can be made by updating the microprogram.
Cost	Higher cost due to complex hardware.	Lower cost due to simpler hardware and programmable nature.
Development Time	Longer development time for physical design.	Shorter development time for programming.
Scalability	Difficult to scale for complex instructions.	Easier to scale by adding more microprograms.

> **Must-Write Points (for 10 marks)**

> **Quick Recall**

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